

Patient Analysis-

Objective

The objective of this analysis is to understand the hospital's patient demographics, registration patterns, and appointment utilization. The analysis aims to identify patient characteristics, evaluate engagement levels, and establish key performance indicators (KPIs) that support operational planning and healthcare decision-making.

Analysis Performed

The following analyses were conducted using Python and Pandas:

- Total registered patients
- New patient registrations
- Gender distribution
- Blood group distribution
- Appointment frequency per patient
- Repeat patient identification

Key Performance Indicators (KPIs)

KPI	Value
Total Patients	30
New Patients	30
Male Patients	15
Female Patients	15
Repeat Patients	20

Results

- The hospital maintains records for **30 unique registered patients**.
- All patients were registered during **June 2026**, indicating that the dataset represents a single registration period.
- The patient population is evenly distributed, comprising **15 male** and **15 female** patients.
- All patient records contain a documented blood group, with **O+** recorded for every patient in the sample dataset.
- Appointment analysis shows that **20 patients have two recorded appointments**, indicating repeat utilization of hospital services.

Key Findings

- The hospital maintains a complete dataset of **30 registered patients**.
- The patient population is equally distributed between male and female patients.
- All patients were registered within the same month, providing a consistent baseline for operational analysis.
- Multiple patients returned for follow-up consultations, demonstrating continued engagement with hospital services.
- Patient appointment distribution is balanced, making the dataset suitable for further analysis of doctor workload, appointment efficiency, and revenue generation.

Business Insights

The patient dataset provides a reliable demographic foundation for hospital operations. The balanced gender distribution and consistent appointment frequency indicate a well-structured sample dataset that supports meaningful operational analysis.

The presence of repeat patients reflects ongoing patient engagement and continuity of care, while complete demographic information improves the reliability of healthcare reporting. These insights can assist hospital management in understanding patient utilization patterns, forecasting service demand, and planning future resource allocation.

Furthermore, the calculated patient KPIs establish the foundation for subsequent analyses, including doctor performance evaluation, appointment efficiency assessment, and revenue analytics.

Conclusion

Patient analysis demonstrates that the hospital maintains a complete and consistent patient database suitable for healthcare analytics. The calculated KPIs and demographic insights provide a strong baseline for evaluating hospital performance and support data-driven decision-making. These findings also serve as key inputs for the Power BI dashboard, enabling effective visualization of patient trends and operational performance.